SHRI G. S. INSTITUTE OF TECHNOLOGY AND SCIENCE

Indore

Student Attendance and Academic Performance Analysis

Real-Life Data Visualization Project

Submitted by
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Tools
Microsoft Power BI implementation kit
Python, Pandas and Matplotlib
Excel-compatible CSV data and offline interactive HTML dashboard

Submission date
23 October 2026

Public-data prototype using UCI Student Performance. The records describe two Portuguese secondary schools and are not actual SGSITS student data. Absence counts are used because attendance percentages are unavailable.

Abstract and report guide

This project examines associations between recorded absences, reported weekly study time and academic grades using the publicly available UCI Student Performance dataset. The Mathematics cohort contains 395 records and the Portuguese cohort contains 649. We analyze each subject separately because some students occur in both files. The default dashboard view uses Mathematics, whose mean final grade is 10.42/20 and whose pass percentage is 67.09% under the project threshold of 10/20.

The full Mathematics cohort shows almost no linear absence-grade relationship (Pearson r = 0.034). The second-period grade is much more strongly associated with the final grade (r = 0.905). These descriptive results support a cautious discussion of early academic support, not a claim that attendance is unimportant or that study time causes higher marks. Valid zero grades and high-absence observations remain in the primary analysis.

Deliverables include original and cleaned data, a repeatable Python analysis, nine charts, this report, a presentation with speaker notes, and a four-page offline interactive dashboard. The Power BI kit contains M transformations, DAX measures, a theme and a detailed build guide. It requires execution in Power BI Desktop before a native dashboard can be claimed as verified.

Eight required submission sections

Section	Coverage
1 Problem Statement	Problem, objectives and intended users
2 Data Collection	Source, scope, fields and provenance
3 Data Preprocessing	Quality checks and derived fields
4 Exploratory Data Analysis	Descriptive statistics and analytical approach
5 Data Visualization	Charts with interpretation
6 Interactive Dashboard	Four pages and Power BI implementation
7 Insights and Recommendations	Findings, actions and limitations
8 Final Submission	Package, reproduction, conclusion and references

1 Problem Statement

Introduction

Academic teams need understandable evidence about patterns in student outcomes. An interactive report can show where grades vary, which groups may need support and whether earlier assessments provide useful warning signals. The college assignment proposes attendance, internal marks, assignment completion and semester results as relevant indicators. This project implements the supported portions with a public dataset and identifies the fields needed for a future campus version.

Problem definition

How are absence counts, reported study time and previous-period grades associated with final academic performance in the available student cohorts? How can these relationships be presented clearly enough for a teacher or academic coordinator to inspect individual groups without overstating causality?

Objectives

- Describe final-grade distributions and the proportion meeting a transparent pass threshold.
- Compare grade outcomes across absence and study-time groups while retaining sample sizes.
- Measure the strength and direction of associations with G3 using Pearson correlation and selected rank-based checks.
- Provide slicers, tooltips and a student-record explorer with drill-down navigation.
- Document missing institutional indicators and define a responsible path to an authorized SGSITS implementation.

Scope and intended use

The analysis is descriptive and exploratory. It does not predict future results, identify causes, rank teachers or assign disciplinary risk scores. G1 and G2 are source-school period grades, not SGSITS internal marks. G3 is the source final grade, not an SGSITS semester GPA. The project title follows the assignment, while every attendance-related visual uses the supported term absences.

2 Data Collection

We downloaded the official Student Performance archive from the UCI Machine Learning Repository on 23 October 2026. The package includes student-mat.csv and student-por.csv, a field dictionary and an R example for matching overlapping profiles. The source documentation describes school reports and questionnaires from two Portuguese schools. Dataset credit: Paulo Cortez, Student Performance, DOI 10.24432/C5TG7T, CC BY 4.0.

Property	Mathematics	Portuguese
Original file in package	Original_Dataset.csv	Original_Portuguese.csv
Course records	395	649
Original columns	33	33
Schools	GP and MS	GP and MS
Final-grade scale	0-20	0-20
Collection dates in CSV	Not present	Not present

The original CSVs preserve the downloaded bytes and semicolon delimiter. The 1,044 appended rows in All_Subject_Records.csv are course records, not 1,044 distinct students. UCI documents 382 overlapping students. We do not create cross-subject person identifiers or assume that demographic profile matching is an error-free identity key. The subject selector therefore permits one cohort at a time.

Fields used in the analysis

Field	Meaning and interpretation
absences	Recorded absence count; scheduled sessions are unavailable
G1, G2, G3	First-period, second-period and final grades on a 0-20 scale
studytime	Ordinal weekly category: <2, 2-5, 5-10 or >10 hours
school, sex, age	Recorded cohort attributes for filtering
failures	Coded previous class failures, interpreted cautiously

The CSV files do not provide calendar dates or a reliable semester time series. The repository donation date is 2014 and the cited source study is from 2008; neither establishes record-level observation dates. Attendance percentage and assignment-completion rate cannot be computed from these files.

3 Data Preprocessing

Analysis.py loads the two original files with a semicolon delimiter and processes each subject independently. It trims categorical text, checks blank and null cells, converts documented numeric fields to integers, validates allowed ranges and categories, and checks exact repeated profiles before creating analytical fields. The shipped data pass every domain check.

Check	Mathematics	Portuguese
Rows before and after	395 / 395	649 / 649
Missing cells	0	0
Exact duplicate rows	0	0
Invalid domain values	0	0
Absence IQR upper fence	20	15
Flagged high-absence records retained	15	21
Zero final grades retained	38	15

Missing values require investigation rather than silent grade imputation. The script raises a clear error if missing, fractional integer-field, invalid-category or invalid-range values appear on a later refresh. Exact repeated profiles also trigger review because the source has no stable person identifier. No rows require deletion in this version.

We define high-absence flags using Q3 + 1.5 × (Q3 - Q1), calculated separately by subject. These are screening flags, not proof of erroneous data. The values lie inside the documented domain, so we retain them without winsorization or replacement. Zero final grades are valid source values and remain in every primary result.

Derived fields

Record_ID combines a subject prefix with the original row position. Pass_Status is Pass when G3 is at least 10. Performance categories are 0-9, 10-11, 12-13, 14-15 and 16-20. Absence groups are 0, 1-5, 6-10 and 11+. Study_Time_Category decodes the source studytime field. Grade_Change equals G3 minus G1. These are transparent project definitions rather than SGSITS academic rules.

4 Exploratory Data Analysis

The unit of analysis is one student-course record. Statistics are calculated within subject, with Mathematics as the primary cohort. Counts, means, medians, sample standard deviations and proportions establish the baseline before relationships are interpreted. The cleaned files retain all original features to support reproducible inspection.

Statistic	Mathematics	Portuguese
Records	395	649
Mean G3	10.4152	11.9060
Median G3	11	12
Sample standard deviation G3	4.5814	3.2307
Pass / fail	265 / 130	549 / 100
Pass percentage	67.09%	84.59%
Mean absences	5.71	3.66
Median / maximum absences	4 / 75	2 / 32

Calculation methods

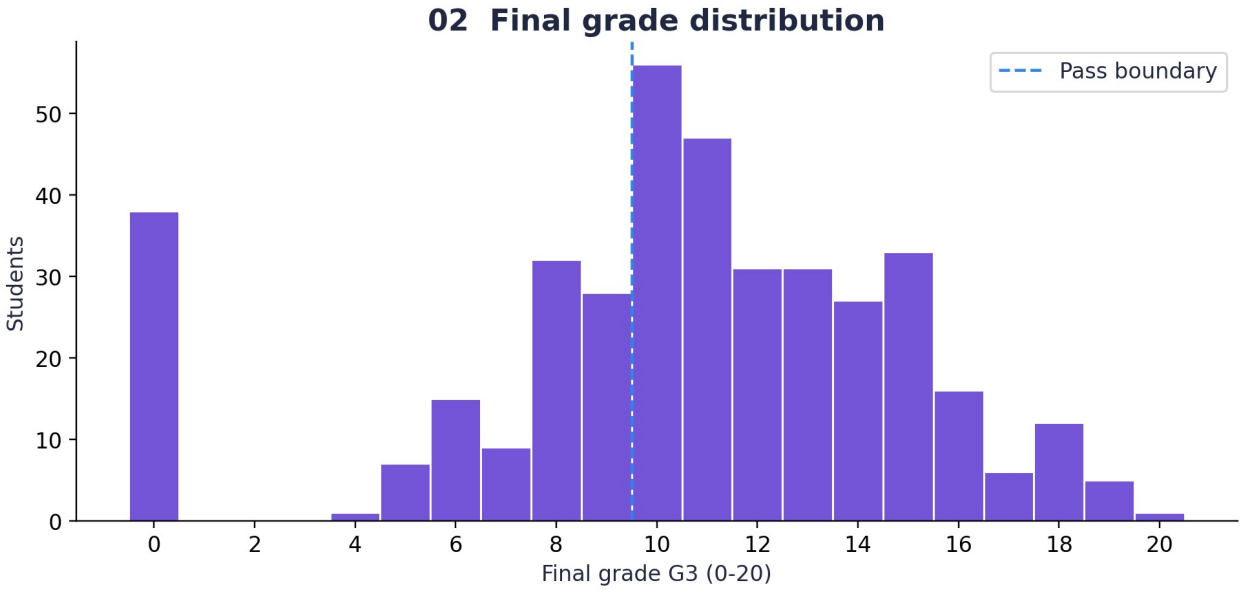
Average marks = sum of G3 divided by the number of selected records. Pass percentage = 100 × number of selected records with G3 ≥ 10 divided by selected record count. Pearson r divides the sum of centered cross-products by the square root of the two centered sums of squares. A zero-variance or insufficient-data selection produces an unavailable correlation, not zero.

Pearson correlation summarizes a linear association. Spearman correlation, implemented as Pearson correlation of ranks, provides a supplementary monotonic check. Study time and prior failures are ordinal codes, so their correlations are exploratory. Scatter plots and group sample sizes accompany coefficients to expose distributional features that a single number can hide.

We report descriptive differences without causal claims, significance tests or population-wide confidence statements. These historical school cohorts are not a random sample of SGSITS students. Filtering to a small subgroup can make a coefficient unstable or undefined.

5 Data Visualization

Final grade distribution



Source: UCI Student Performance | Mathematics unless otherwise noted | Grades 0-20

Figure 1. Mathematics final grades using one-point histogram bins.

The Mathematics cohort has 265 passing and 130 failing records under G3 ≥ 10, giving a pass percentage of 67.09%. The mean is 10.42, below the median of 11. The distinct group of 38 zero final grades affects the overall distribution and is retained as observed.

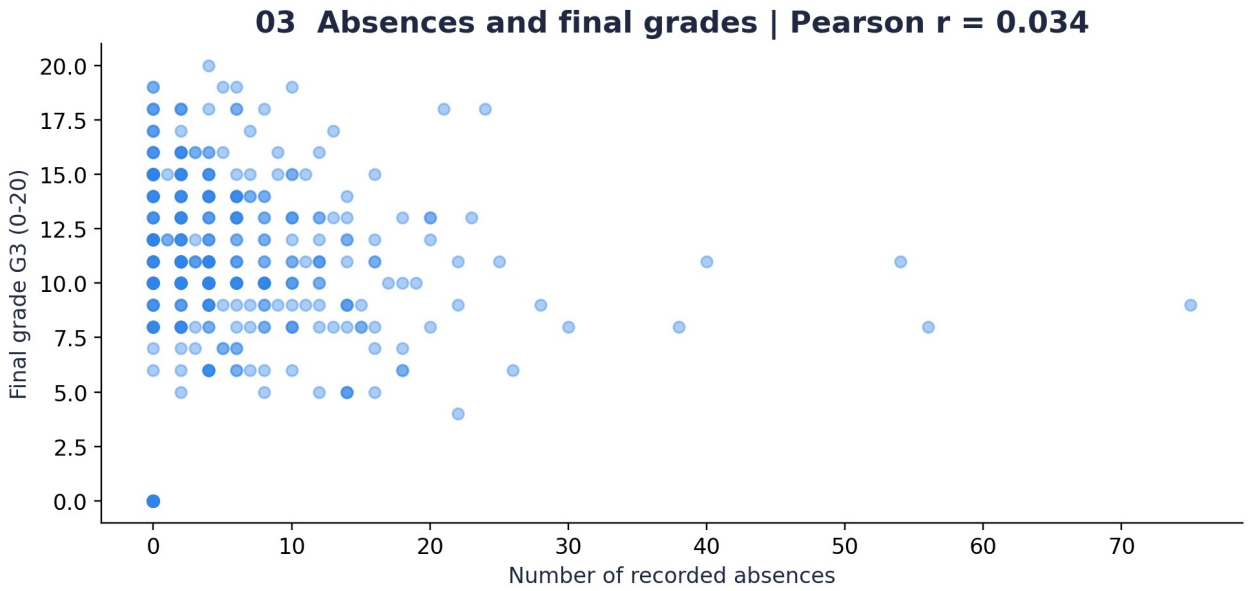
A histogram is appropriate because grades are numeric and discrete. One-point bins preserve the original scale and avoid hiding the pass boundary in a wide interval. The separate pass/fail chart in the chart folder and dashboard summarizes the binary project rule, while the histogram shows variation within each outcome.

Interpretive limit

The final-grade field does not explain why a student received zero. It would be unsupported to label every zero as a dropout, absence from the examination or a missing result. Any institutional investigation would need additional authorized records.

5 Data Visualization continued

Absences and final marks



Source: UCI Student Performance | Mathematics unless otherwise noted | Grades 0-20

Figure 2. Mathematics absence counts against final grades, with all 395 records.

The full-cohort Pearson correlation is 0.034, with Spearman correlation 0.018. Both are close to zero. The scatter contains repeated integer coordinates, so transparent markers reduce overplotting, although identical records can still visually overlap. The interactive alternative exposes record details on hover and click.

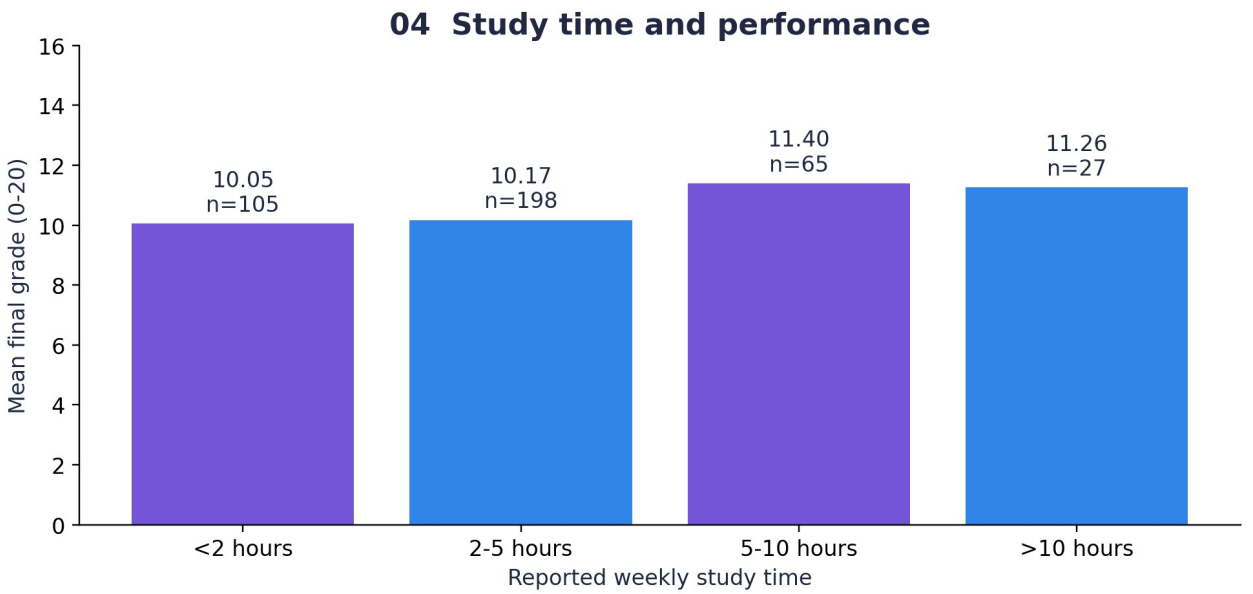
This result does not demonstrate that attendance has no educational importance. The dataset measures recorded absences rather than a standardized attendance proportion, and unobserved circumstances may influence both absences and grades. There is no total-session denominator with which to compare attendance exposure.

Group comparison

Mathematics mean G3 is 8.44 for zero absences (n=115), 11.65 for 1-5 (n=134), 11.40 for 6-10 (n=80), and 10.15 for 11 or more (n=66). The pattern is not monotonic. The supplementary group chart shows this explicitly and prevents a misleading straight-line narrative.

5 Data Visualization continued

Study time and performance



Source: UCI Student Performance | Mathematics unless otherwise noted | Grades 0-20

Figure 3. Mathematics group means with sample sizes for reported weekly study time.

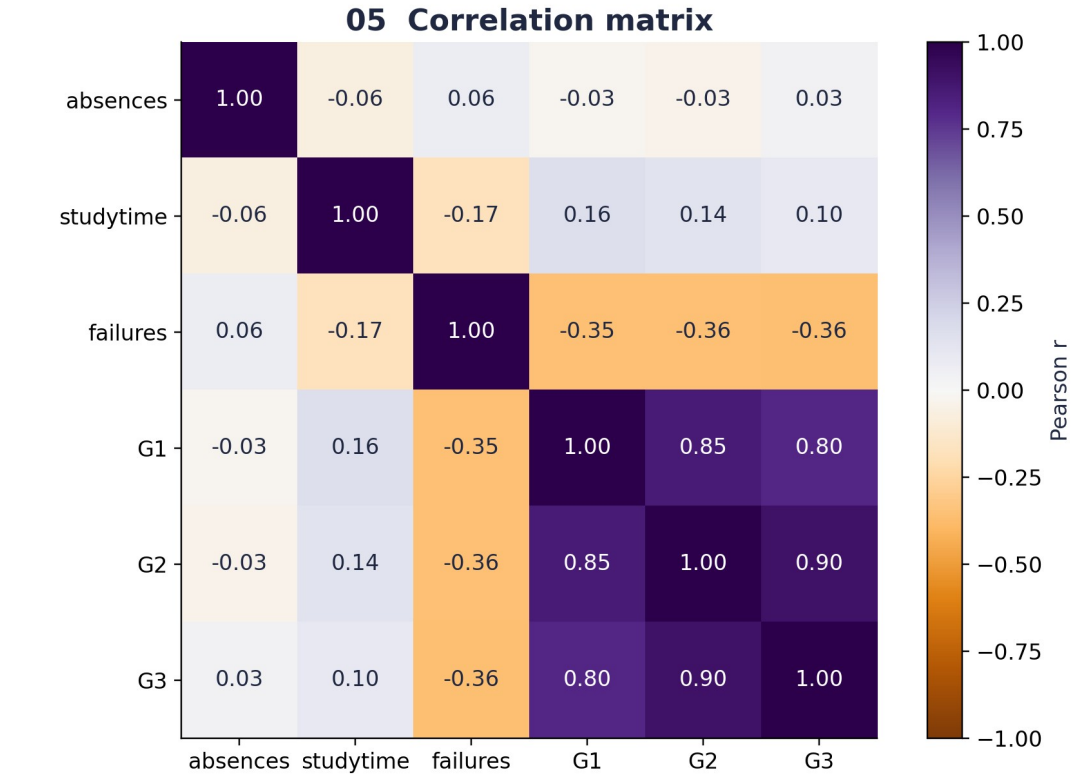
Mean Mathematics grades rise from 10.05 in the under-two-hours group to 11.40 in the five-to-ten-hours group, then decrease slightly to 11.26 in the over-ten-hours group. Their sample sizes are 105, 198, 65 and 27 respectively. The smallest group should not receive the same interpretive weight as the largest.

The coded studytime correlation with Mathematics G3 is 0.098; the rank-based coefficient is 0.105. These are weak associations. The source categories describe reported time bands and are not precise measurements of learning quality, study efficiency or motivation.

The additional box plot in All_Charts compares within-group spread and medians. This helps distinguish a difference in group averages from a separation of individual outcomes. Higher reported study time is a useful discussion prompt, but this analysis cannot estimate the effect of assigning a student an additional hour of study.

5 Data Visualization continued

Correlation analysis



Source: UCI Student Performance | Mathematics unless otherwise noted | Grades 0-20

Figure 4. Pearson correlations in the full Mathematics cohort.

G2 has the strongest displayed association with G3 (r = 0.905), followed by G1 (r = 0.801). Previous-failure codes are negatively associated with final grades (r = -0.360). Absence counts and reported study-time codes have weaker full-cohort relationships.

The matrix uses a fixed -1 to +1 scale and includes numeric annotations. It supports comparison of direction and strength, not causal attribution. G1 and G2 occur earlier in the academic sequence, but G2 would be unavailable to a support system operating before the second assessment. Any predictive extension must respect that timing to avoid information leakage.

5 Data Visualization continued

Previous grades and final grades

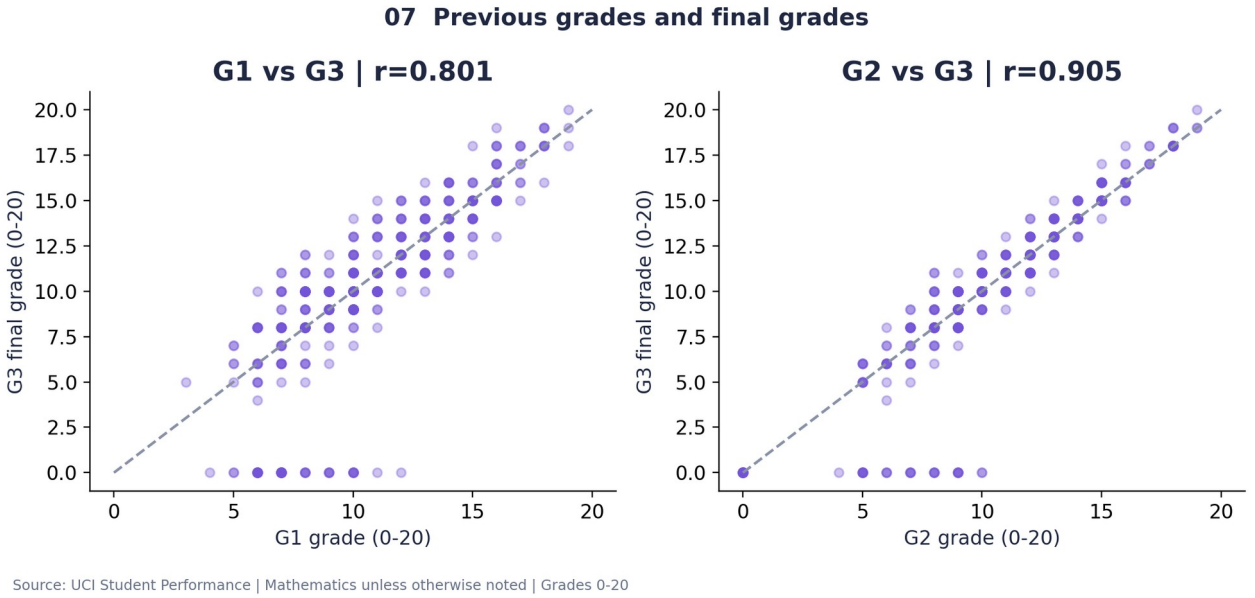


Figure 5. First-period and second-period grades against final Mathematics grades.

Both scatter plots show strong positive associations, with G2 more closely aligned with G3. The dashed diagonal marks equal earlier and final grades. Observations below that reference have a lower final grade than the earlier grade; those above it have improved on that scale.

A practical interpretation is to review academic progress after each assessment and offer support where performance remains low or falls. This is a proposed use of the pattern, not evidence that a particular intervention would improve grades. Individual cases should include contextual discussion with the student.

Why there is no semester trend line

G1, G2 and G3 identify assessment periods, but the files lack record-level calendar dates and an SGSITS semester sequence. We therefore use paired grade comparisons and do not invent dates, semesters or a historical time series. A future institutional dashboard could add a genuine trend only after authorized, time-stamped records become available.

5 Data Visualization continued

Subject cohort comparison

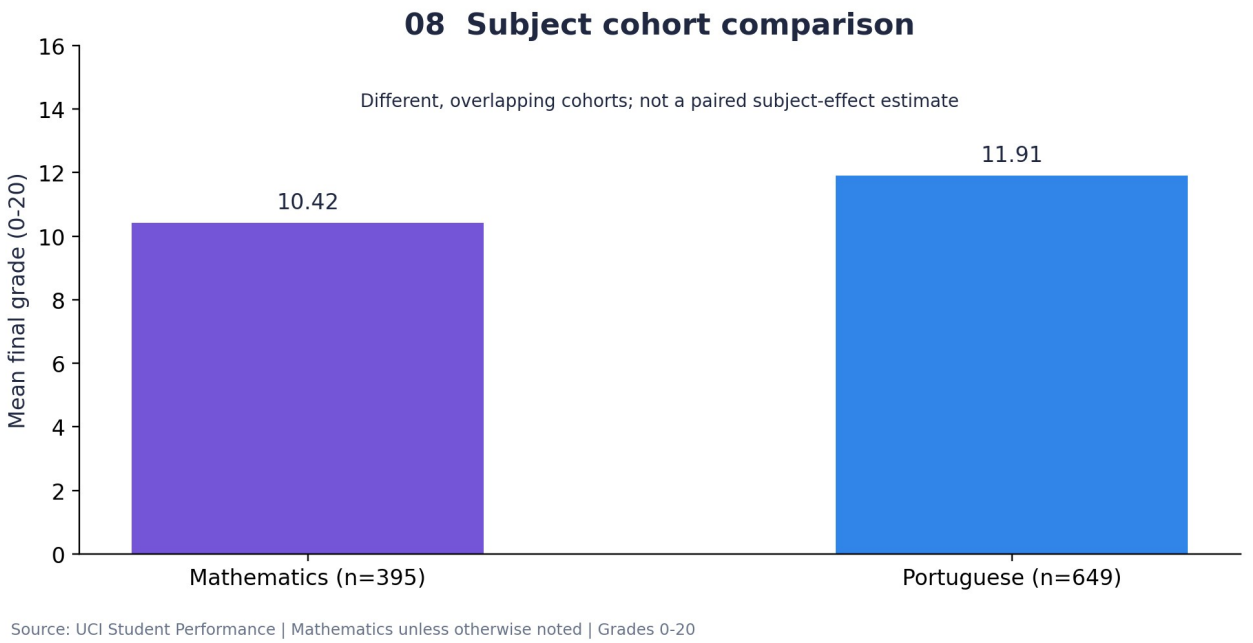


Figure 6. Cohort mean G3 for Mathematics and Portuguese, with different cohort sizes.

Portuguese has mean G3 11.91/20 and a pass percentage of 84.59%, compared with 10.42/20 and 67.09% in Mathematics. The unadjusted mean difference is 1.49 grade points. These are descriptive comparisons across overlapping but unequal cohorts.

The comparison is not a within-person estimate of a subject effect. It does not establish that one subject is easier, that one teaching approach is better, or that the difference would recur at SGSITS. The dashboard comparison chart ignores only the subject selector and retains the other filters, with sample sizes in the tooltip or labels.

Design consistency

Purple and blue distinguish series throughout the report, slides and dashboard. White plotting areas, readable axes and explicit units keep the charts consistent. All nine PNG charts are generated from the same cleaned data and are included at 220 DPI for reuse. Their calculation source is Analysis.py.

6 Interactive Dashboard

The delivered dashboard is a self-contained HTML alternative with embedded data, JavaScript and SVG charts. Open Interactive_Dashboard.html directly in a modern browser after extracting the ZIP. It works offline and uses no external chart libraries or remote data requests. The interface explicitly identifies the public-data scope and the absence of an attendance percentage.

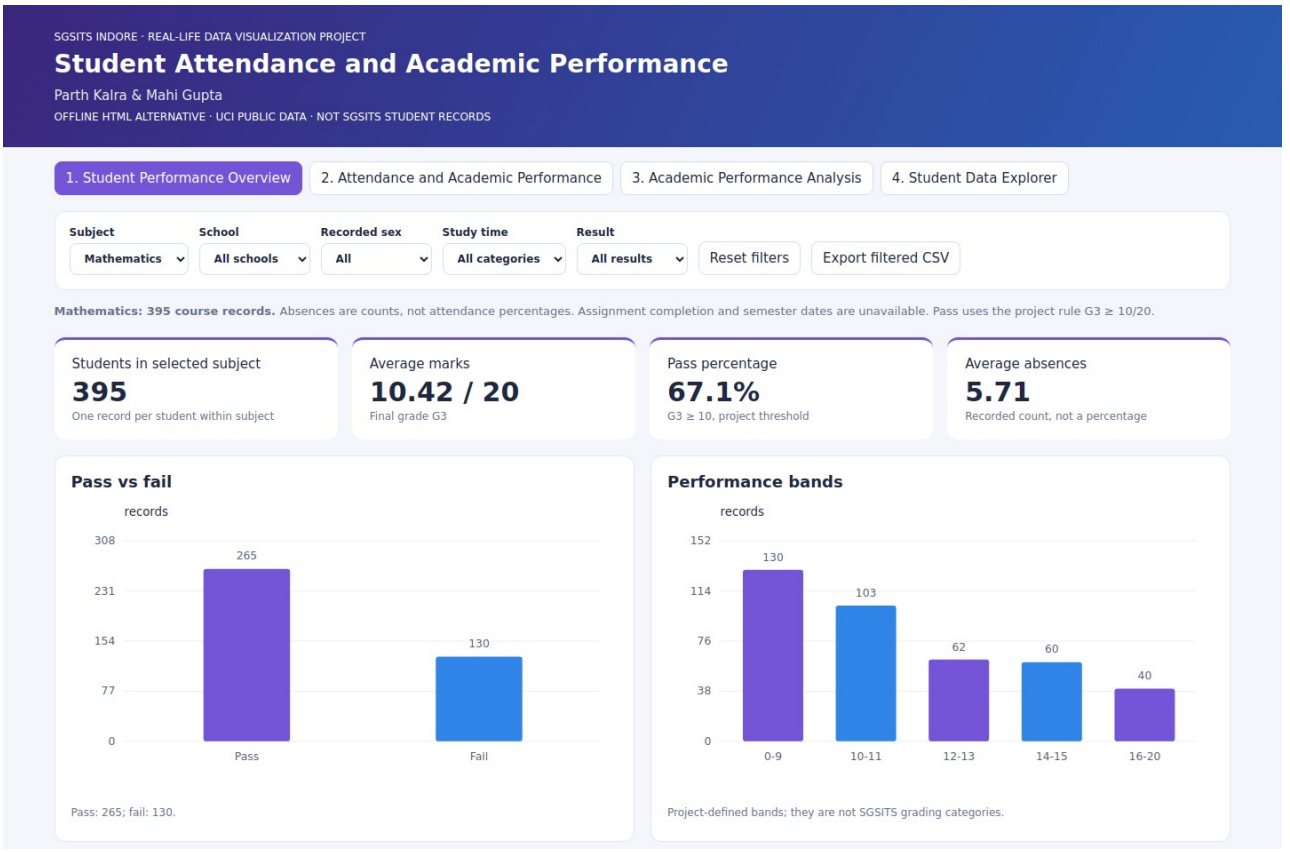


Figure 7. Implemented offline dashboard overview. This is not a Power BI screenshot.

Page	User task
Student Performance Overview	Inspect KPIs, pass/fail counts and grade bands
Attendance and Academic Performance	Explore absences, study time, correlations and absence groups
Academic Performance Analysis	Compare earlier grades, grade distributions and subject cohorts
Student Data Explorer	Search, sort and inspect records; expand school and study-time groups

6 Interactive Dashboard implementation

Interactions and interpretation

Subject, school, recorded sex, study-time category and result filters update the cards and charts. The subject filter is single-select, so a student count always refers to one subject. Hovering over chart marks shows values. Clicking a scatter point or record ID opens a detailed anonymous course record. The explorer supports column sorting, ID search and school-to-study-group drill-down. Filtered records can be exported as CSV.

The explorer search narrows only the displayed table. The top-level slicers define KPI and export scope. A subject comparison on page 3 deliberately shows both cohorts while keeping the other slicers. Empty selections show no records, and undefined correlations show N/A.

Power BI model and reproducibility kit

PowerQuery.m loads each raw CSV, validates fields, creates derived columns and appends the subject records into Students. A disconnected SubjectAxis table enables the comparative chart. DAX_Measures.dax contains the KPI definitions and filter-aware correlation calculations. Theme.json supplies the color palette and Page_Blueprint.json specifies visual field assignments. The setup PDF describes each step, tooltip configuration, sort orders and drill-down hierarchy.

Power BI Desktop was unavailable in the build environment. The package therefore contains no native PBIX or PBIP. The M and DAX files have been reviewed against source definitions and reference formulas but were not executed in a Microsoft engine. The blueprint is a design specification and must not be opened or renamed as a Power BI project. Complete the supplied acceptance checks in Desktop before calling the native report verified.

Layout specification

Use a 1280 × 720 canvas, purple and blue series, a pale background and white visual surfaces. Keep the four named analysis pages. Synchronize slicers and add Record_ID, school, G1, G2, G3 and absences to scatter tooltips. Add school, Study_Time_Category and Record_ID as a matrix hierarchy on the explorer page. Use the exportable PNGs as presentation assets, not as substitutes for interactive Power BI visuals.

7 Insights and Recommendations

Evidence-based findings

1. Previous grades show a stronger relationship with final Mathematics performance than absence counts in the full cohort. G2-G3 r is 0.905 and G1-G3 r is 0.801, while absences-G3 r is 0.034. This supports considering earlier assessments in a descriptive academic review.

2. Mathematics has 130 results below the project pass threshold, representing 32.91% of its cohort. The corresponding Portuguese figure is 100 results or 15.41%. These provide cohort baselines, not predictions for individual students or SGSITS.

3. Study-time group averages differ modestly in Mathematics. The five-to-ten-hours group averages 11.40 compared with 10.05 for under two hours, an unadjusted difference of 1.35 points. The over-ten-hours group averages 11.26 and contains only 27 records, so a simple more-hours-always-means-higher-marks interpretation would be inaccurate.

4. Subject differences depend on the population being compared. Portuguese has a 1.49-point higher unadjusted mean, but the cohorts overlap and differ in size and composition. A paired or adjusted analysis would require carefully justified identity linkage and model assumptions.

Sensitivity check for zero grades

The main Mathematics analysis retains all 395 records. In a separate sensitivity subset containing only G3 > 0, the sample decreases to 357 and the absence-grade correlation becomes -0.213. This is a different population and is not a corrected or preferred estimate. Excluding records according to the outcome can introduce selection bias. The change demonstrates why data definitions and sensitivity checks matter.

The Portuguese absence-grade coefficient is -0.091 in the full cohort and -0.196 in the positive-grade subset. Again, this does not justify removing zeros. All primary charts, KPIs and cleaned datasets retain them.

7 Recommendations and limitations

Practical recommendations

Use earlier assessment results as a discussion aid for timely support. A department could review low or declining marks after each assessment, ask students about learning barriers and offer tutoring or mentoring. This project does not establish an optimal intervention threshold or prove that such support improves outcomes.

Combine academic review with valid attendance measurements. Record sessions attended and sessions scheduled for the same student, course and period before calculating 100 × attended / scheduled. Handle cancelled classes and excused absences through a documented policy, and avoid comparisons across incompatible denominators.

Collect assignment evidence only through authorized systems. Completion rate requires assignments completed and assignments assigned, with a clear treatment of exemptions and late submissions. Store assessment dates and a documented student-course key so future analyses can follow actual semester progression.

Evaluate any pilot before wider use. Compare outcomes using an appropriate study design, monitor the size of each subgroup and seek student feedback. Provide explanations and human review rather than automated disciplinary labels based on correlations.

Limitations

The data represent historical Portuguese secondary education, so external validity for a contemporary Indian engineering college is limited. The analysis is observational and cannot separate causal effects from confounding, reverse relationships or self-report error. Studytime is ordinal; it does not measure study quality. Absence denominators, assignment records and observation dates are missing.

Subject cohorts overlap, but no stable cross-subject identifier is supplied. Course-row counts cannot be treated as distinct-student totals. Small filtered groups can produce unstable associations. Valid zero grades and extreme absence counts affect summary patterns. The Power BI code still requires local execution, and the presentation has not been tested inside Microsoft PowerPoint.

Privacy and institutional use

The delivered Record_ID is a generated course-row label. An SGSITS deployment should use approved, minimized and anonymized data, restrict access to individual records and publish aggregate results where possible. Public cohort patterns must not be attributed to named SGSITS students.

8 Final Submission

Deliverable	Location and use
Data	01_Dataset: untouched sources, cleaned cohorts, appended course records, source dictionary
Power BI kit	02_PowerBI: M, DAX, theme, page specification and setup PDF
Analysis	03_Python: runnable script, dependencies, statistics and audit logs
Charts	04_Visualizations/All_Charts: nine PNG visualizations
Report	05_Report: editable Word and matching PDF
Presentation	06_Presentation: 18-slide PPTX with speaker notes
Interactive alternative	07_Dashboard_Alternative: standalone HTML
Verification	Verification.md and SHA256SUMS.txt at project root

Reproduction

Extract the entire ZIP before opening files. Install Python 3.10 or later, create a virtual environment, install 03_Python/requirements.txt, and run python 03_Python/Analysis.py from the project folder. The script resolves paths relative to itself and regenerates cleaned data, statistical outputs and charts without downloading anything. A normal modern browser can open the HTML directly. The Power BI setup guide explains the separate Desktop build.

Conclusion

The project delivers a reproducible exploration of absence counts and academic performance while preserving the limitations of the public source. Previous grades are strongly associated with final results, whereas the full-cohort Mathematics absence relationship is near zero. Study-time and subject comparisons show descriptive differences that require careful interpretation. The dashboard makes these patterns inspectable and provides a practical foundation for a future, authorized SGSITS analysis with genuine attendance and assignment data.

The principal remaining implementation step is native Power BI execution and acceptance testing on a Windows machine with Power BI Desktop. The working offline HTML alternative is separately labeled so the submitted materials do not misrepresent native Power BI validation.

References

College assignment

Real-Life Data Visualization Projects (1)(3).pdf, provided assignment. Project 1 and common eight-step submission requirements.

Dataset

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T

Repository

https://archive.ics.uci.edu/dataset/320/student+performance

Original download

https://archive.ics.uci.edu/static/public/320/student+performance.zip

License

https://creativecommons.org/licenses/by/4.0/

Power Query CSV

https://learn.microsoft.com/en-us/powerquery-m/csv-document

DAX filter context

https://learn.microsoft.com/en-us/dax/calculate-function-dax

Power BI themes

https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-report-themes

Power BI projects

https://learn.microsoft.com/en-us/power-bi/developer/projects/projects-overview

Source access and download date: 23 October 2026. All numerical findings in this report were calculated by the supplied Analysis.py. The downloaded UCI dictionary and source hashes are included for provenance. URLs are provided in full for copying and source verification.